

system fails. Web sites currently do not record biometrics; the bundled software will send passwords to the site when you are authenticated via biometrics. So the need for strong passwords doesn't go away. There are dozens of such systems available, and all perform with varying levels of success. Devices like the Sony Notebook Camera Kit let you access the PC only if you show your friendly face and the DigitalPersona U.are.U Pro will be happy to see your clean fingers.

Could it be magic?

Biometrics is essentially the measurement of physiological or behavioural characteristics of a person. It is important to measure biological characteristics that are unique to individuals, difficult to fake, and easy to measure. On this basis, fingerprints, handprints, voice and face top the easy to measure criteria, while iris and retina scans provide the highest level of unique, hard to fake biometrics. These are all physical characteristics of your body, and therefore a part of who you are. You can be reasonably sure (odds of 1 to a million) that you



- Most convenient, great for public places.
- Quite inaccurate. Is affected by changes in lighting, age, posture, movement, glasses, etc.

won't bump into another person with the same characteristics. This is what biometric security banks upon.

Biometrics does not guarantee security. It simply offers a more convenient and reliable means of identification and authentication. Identification is a process of recognition and involves taking the biometric data and searching for a match in the database, making it a slower process that is more prone to errors. Authentication involves confirming your identity, and the biometric system has to match your data with only one, single record.

Some systems measure things you do, in addition to the things you are. When you type your password to login to your computer, you will notice that your typing has a certain pattern to it. Some letters follow others in quick succession, while others take longer. This pattern is almost a ritual that is very closely repeated every time you type, and is also reasonably unique to you. A

Do I know You?

It is quite easy to procure anyone's fingerprints, so manufacturers take steep measures to ensure that the prints are only scanned from a live finger and not off copies. A Yokohama University professor, Tsutomu Matsumoto recently published his techniques for fooling fingerprint scanners. He found that optical fingerprint scanners can often be fooled by silicone rubber fingers. However, electromagnetic scanners would reject those because their electrical characteristics are different from that of a live finger. He didn't have to look far for a suitable material to simulate a live finger. Jelly sweets, made of gelatine, were found to have just the right electrical properties and a surface that is easy to mould. Sheets of clear gelatine with the fingerprint impressed, wrapped around your finger, would be enough to fool the scanner even in the presence of a security guard. Matsumoto was able to consistently fool eleven commercially available systems 80 per cent of the time. With complete details on transferring prints from any surface to gelatine sheets now publicly available, this

development makes a huge dent in the trustworthiness of fingerprint systems.

In public locations such as banks and airports a large number of people transit every day, including known criminals. Face recognition systems attached to video surveillance systems are supposed to help catch them. Iceland has already installed a face recognition system sold by Visionics to improve security. Theoretically it is an excellent idea, but experiences indicate otherwise. The Visionics system was tested for eight weeks at Palm Beach airport in Florida. From all the people passing by the surveillance systems, it could identify less than half of the 15 people registered with it. It was adversely affected by lighting, movement and people wearing spectacles. Law commentator Henry Miller calls this the 'Clark Kent syndrome'. With spectacles it's Clark Kent, take them off, and you have Superman!

One vendor, FaceKey, says candidly that face recognition is practically useless for general identification, but ideal for individual authentication.

software called BioPassword from Net Nanny, retailing for less than \$100 (approx Rs 5,000), integrates with Windows 2000 and Windows NT login to provide this authentication, without the need for any additional hardware.

Depending upon the technique used, biometrics have varying levels of accuracy. Moreover, since the measurements here are of actual people, there can be no predictability of response. The system might let you pass in one attempt on Monday and require five on Tuesday. Biometric systems are unique in the matter of having False Acceptance Rates (FAR)—the probability of the system falsely accepting a metric as a match, and False Rejection Rates (FRR), where the system fails to recognise an authentic bio-signature. Almost all biometrics systems can be adjusted to varying levels of 'strictness' to create a variation in FAR and FRR. Obviously, as the FRR is increased, the FAR goes down, and vice-versa.

There are no accepted standards for

measuring the accuracy of biometrics on both these criteria, so we often have to rely on manufacturer's own data or the results of studies that often do not agree on the results. Yet, there is a broad consensus on the relative accuracy levels of different biometric techniques. One study sponsored by the British Communications Electronics Security Group last year showed Iris-based identification systems had no false matches in 2.7 million trials, with a false rejection rate of less than 2 per cent in first attempts and zero in three attempts. Face recognition showed false matches of over 10 per cent, and fingerprint systems had false matches of less than 0.5 per cent (see box 'Do I know You').



- Cheap and easy to deploy, convenient for personal use.
- People often lack readable fingerprints, extremely easy to fool.

Common biometrics

The idea of a computer looking at a face and recognising it is inherently enticing. It is the least intrusive method of biometrics. Your face is permanently on display no matter where you go, and

